SpectraSense Demo Video

Simple spoken script + screen plan | about 4 min 45 s | revised 18 Oct, 21:30 · highlighted = changed or new

Before you record (10 minutes)

- Backend and frontend must be running. Open **localhost:8000/health**: it must show **"llm": "groq"**. Do one full test run (H04). In Logfire, the trace of that run should show **llm assess**; if a run took less than a second or the report looks short, the large model was busy: wait 10 minutes and try again.
- **Observability is verified:** Logfire shows each request, the safety gate and every model call with its time. LangSmith shows the pipeline steps but **no model calls or tokens**: if you show LangSmith, show only the steps.
- Open these tabs in this order and switch with Ctrl+Tab:
  1. Home page (localhost:3000)
  2. Assess page with H04 ready: **localhost:3000/assess?example=H04**
  3. A second Assess tab, empty (for the Karl Fischer question)
  4. Logfire: Live view, filtered to the last 15 minutes
  5. Architecture slide: **docs/video/architecture.png** in the repository (open it full screen)
  6. Evals page (localhost:3000/evals)
  7. GitHub: github.com/yashwanth-1341/spectroscopy
- Copy the Karl Fischer text to your clipboard: "We check water in transformer oil with Karl Fischer titration. We want an inline NIR probe so we can stop the titration. NIR needs no calibration, right?"
- Do **not** press Run evals before recording: the Evals page already shows tonight's judge results (marked 18 Oct, 21:33), and a new run would use the model quota you need for the live demo.
- Browser zoom 125%. Close chat apps, mute notifications, hide bookmarks.
- Record a spare take of the H04 run and the Karl Fischer run. If a live run is slow or odd, cut to the spare.

How to sound clear and warm

- Read slowly. If a sentence feels too slow, you are about right.
- Stop at every [pause] and take one breath.
- Smile when you say the hook and the last line. People can hear it.
- Say numbers slowly: "two point four micrometres", "ninety-seven percent".
- If the screen shows slightly different words from this script, describe what you see. Do not read words that are not there.

1. Hook (0:00 to 0:30)

SHOW ON SCREEN	SAY (read this out)
1. Home page. Keep the mouse still. 2. On the word "SpectraSense", move the mouse slowly to **Start an assessment**. Do not click yet.	Imagine you work in a factory. You know what you want to measure. Maybe water in oil. Maybe the amount of medicine in a powder. [pause] But you do not know how. Which method? Which light range? Which cable and probe? [pause] Today, you need an expert to answer that. And experts are busy. [pause] So we built SpectraSense. It writes a first draft of that answer in seconds. It only uses the AFS documents. And it shows the proof for every claim.

2. A hard case from the organisers: H04 (0:30 to 1:35)

SHOW ON SCREEN	SAY (read this out)
1. Switch to the H04 tab (the enquiry is already in the box). Click Send. 2. While it runs, point at the progress steps in the chat panel. 3. When the report appears, put the mouse on the **Answer**: "No". 4. Slowly move down: the premise check with the fibre names, the line about the distance, then the questions.	Let us try a hard case from the organisers. [pause] A customer wants to measure water in a petroleum pipe. They want to use mid-infrared light. Their machine is in another building. And they want to reuse an old glass fibre cable. They ask: is this okay? [pause] While it works, it sorts what the customer really said from what we still do not know. And it finds the right pages in the AFS documents. [pause] Now look. The first line is the answer: No. [pause] Why? Glass fibre only carries light up to about two point four micrometres. Mid-infrared starts at two point five. So the signal cannot get through. And it points to the AFS slide that says so. [pause] But it does not stop at no. It lists the cables that do work: chalcogenide, polycrystalline infrared, and hollow waveguide. It says the long distance to the other building is not shown in the AFS material. And it only asks the customer for things the customer can really tell us. [pause] This check is not the AI's opinion. It is a rule, built from the fibre ranges in the documents.

3. Proof for every claim (1:35 to 2:15)

SHOW ON SCREEN	SAY (read this out)
1. Click one citation chip next to the Answer. 2. A small box opens : document name, page number, exact sentence. 3. Point at the line **"Quote found on the cited page"** . 4. Close it. Optionally click a second chip.	Every claim has one of these small tags. [pause] Click it. You see the document, the page, and the exact sentence. And this line says the quote was found on that page. [pause] The AI cannot make up a quote, because it picks passages by number. Every number in the report comes from the customer or from a cited page. If not, it says: "to be agreed with the customer". And a customer fact is kept only if those are the customer's own words. Anything else is marked as an assumption. [pause] So if an expert disagrees, they do not argue with a black box. They click, read the source, and decide.

4. A new question it has never seen (2:15 to 3:00)

SHOW ON SCREEN	SAY (read this out)
1. Switch to the empty Assess tab. 2. Paste the Karl Fischer text. Click Send. 3. When it is done, point at the Answer and at the premise check **"A calibration and the reference method are still needed"**. 4. Scroll to **First feasibility experiment**. Move the mouse down the steps.	Now something it has never seen. [pause] A customer checks water in transformer oil with a lab method called Karl Fischer titration. They want a probe in the pipe instead. And they think near-infrared needs no calibration. Right? [pause] The assistant catches that hidden guess. The answer is no. Near-infrared needs a calibration against the lab method. The lab test stays until the new method is proven. And it cites the AFS laboratory guide. [pause] Now look at the first experiment. It is a real plan. First, lab samples against the reference. Then a test on batches the model has not seen. Then in the pipe. And, before anyone buys anything, agree with the customer how to decide.

5. How it works inside (3:00 to 3:35)

SHOW ON SCREEN	SAY (read this out)
1. Show **docs/video/architecture.png** full screen. 2. Point at each box as you name it: safety gate, understand, facts and unknowns, find pages, check assumptions, write and verify. 3. Switch to Logfire for three seconds: point at one **POST /assess** row and its **llm assess** and **safety gate** lines with their times. (Verified working.)	Here is how it works inside. First, a safety gate checks every message. Off-topic requests, tricks, and prompt injections are stopped before anything runs. [pause] Then comes a chain of steps. Understand the request. Split facts from unknowns. Find the AFS pages. Check the assumptions. Write the report. And check it again. [pause] The idea is simple: the AI writes the draft, and rules decide what stays. [pause] And we can see every run. In Logfire we see each request, the safety gate, and every model call with how long it took, and when a backup kicks in. So if an answer is wrong, we can find out why.

6. How we tested it (3:35 to 4:15)

SHOW ON SCREEN	SAY (read this out)
1. Open the Evals page. Pause on the four score rings at the top. 2. Scroll to **Every enquiry, every mark**. 3. Scroll to **See for yourself** and open one enquiry: the organisers' useful pages (green = cited) next to the sources.	We did not only test by hand. We graded the assistant's answers to all fourteen organiser enquiries. [pause] Code checks the hard facts. Is every quote on its page? Is every piece of advice sourced? Does it ask what it does not know? Does it reach the right verdict? [pause] A second AI model reads for meaning. Is each statement backed by the pages? Does it answer the question? [pause] Every quote is found on its page, and all fourteen reach the expected verdict. The judge found that about ninety-five percent of the statements it checked are backed by the pages. And ninety-seven percent of the pages the organisers marked as useful reach the assistant. With plain search, it was only thirty-five percent. (If the **Nothing invented** card shows a different number by then, say that number.) [pause] You can check any answer yourself against the organisers' list.

7. Close (4:15 to 4:45)

SHOW ON SCREEN	SAY (read this out)
1. Open the GitHub repository page. 2. Scroll to **For the jury: where everything is** at the top of the README. 3. Open the **answers** folder. Show the files H01 to H06. 4. End on the repository page or the SpectraSense home page. Hold two seconds after "thank you".	Everything is in the public repository. The code. A step-by-step setup for Windows and Mac. And the assistant's own answers for all six final cases, H01 to H06. The assistant wrote them. We did not edit them by hand. [pause] SpectraSense does not replace the expert. It gives them a faster first draft that they can trust and check. And when the proof is not there, it is honest enough to say: "not yet". [pause] Thank you.

Last check before you press record

- Every number you say must be one of these measured ones: **2.4** and **2.5** micrometres (AFS documents), **fourteen** enquiries, **ninety-seven** and **thirty-five** percent (retrieval), **ninety-five** percent (judge: statements backed by the pages, the **Nothing invented** card). Do not add others.
- If the live H04 or Karl Fischer answer is worded differently, describe what is on screen and keep the same flow.
- Judge results on the Evals page: marked on 18 Oct at 21:33 by **gpt-oss-20b**, a different model from the assistant's gpt-oss-120b, on the stored answers (the same files as in **answers/**). 35 of 43 judge marks were done; the rest show as *not marked* because the judge hit its daily limit, and are filled in automatically as its quota frees up. Do not hide them. If asked: "the judge model hit its daily limit; the code checks are complete."